



**Bangladesh Industrial Technical Assistance
Center (BITAC)
ATTACHMENT TRAINING PROGRAM**

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Year/Semester: 2nd Year / 2nd Semester **Group :** ME-5

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Table of Contents

Acknowledgement	2
Introduction	3
Day 01	4
Computerizrd Numerical control Shop(CNC)	4
Welding section	7
Fabrication Section	9
Day 02	10
Tools and Technology Institute (TTI)	10
Automobile Section	12
Day 03	14
Heat treatment(HT)	14
Foundry Shop	15
Pattern shop	18
Day 04	20
Mould	20
Industrial Engineering Division(IED)	21
Production Control Division(PCD)	23
Conclusion	24

Acknowledgement

The authors of this report wish to express their sincere gratitude to the Department of Mechanical and Production Engineering (MPE) at Ahsanullah University of Science and Technology and the Bangladesh Industrial Technical Assistance Centre (BITAC) for providing us with the opportunity to participate in the "Attachment Training Program." We would like to extend our heartfelt appreciation to our supervisor, Mr. Md. Araf Hossan, Lecturer Grade-II, Ahsanullah University of Science and Technology and Mr. Md. Hasib Mahmud, Assistant Engineer (Training Division) at BITAC, for their invaluable guidance, assistance, and unwavering support throughout this publication. We are also grateful to all the assistant engineers who generously offered their assistance during the training program. We would also like to acknowledge the technicians of the workshop, whose cooperation and support allowed us to observe and gain insights into the work processes. Without their cooperation, this report would not have been possible. Finally, we would like to thank everyone who contributed to this report and made it a success.

Introduction

Bangladesh Industrial Technical Assistance Centre otherwise known as BITAC is the successor to Pakistan Industrial Technical Centre (PITAC). It was renamed BITAC after the independence of Bangladesh. BITAC was established in 1962 by merging two other productivity-oriented public sector organizations namely IRDC & PIPS.

BITAC is a significant technical assistance center in Bangladesh that plays a crucial role in the development of the country's industrial sector. Since its inception, BITAC has been committed to promoting economic growth by developing skilled manpower, improving productivity, and fostering innovation in technology.

BITAC provides various technical assistance services to the industrial sector, including training, research, and laboratory and workshop support. It offers customized training programs for industrial workers to enhance their technical skills and improve productivity. The center's research and development activities aim to create innovative solutions to industry-specific problems and promote the adoption of advanced technology in the country's manufacturing sector.

One of BITAC's notable initiatives is the manufacturing and repairing of import alternative equipment. This initiative supports the development of local industries by providing cost-effective alternatives to imported machinery and equipment. It also helps to reduce the country's reliance on imported goods and creates employment opportunities for local workers.

BITAC has five centers located in strategic locations across Bangladesh, including Dhaka, Chittagong, Chandpur, Khulna, and Bogura. Each center is equipped with modern facilities and state-of-the-art equipment to support the center's training, research, and laboratory activities. In summary, BITAC is a critical institution in Bangladesh's industrial sector, playing a vital role in promoting economic growth through the development of skilled manpower, improving productivity, and fostering innovation in technology. The center's commitment to providing technical assistance services to the industrial sector has helped to create a more sustainable and robust economy in the country.

Day 01

CNC Lathe Machine

A **CNC lathe machine** (Computer Numerical Control lathe) is a powerful tool used for precision machining. Let's explore the key aspects of CNC lathes:

1. Definition and Technology:

- A CNC lathe is a computer-controlled machine that automates the turning process.
- It rotates a workpiece while cutting tools remove material to create cylindrical shapes.
- CNC lathes use G-code instructions to control tool movement, spindle speed, and feed rates.

2. CNC Lathes Working Processes:

- The operator programs the desired shape and dimensions into the CNC system.
- The machine automatically positions the cutting tool and workpiece.
- The spindle rotates the workpiece, and the tool removes material to achieve the desired shape.
- CNC lathes can perform various operations, including facing, turning, threading, and grooving.

3. Types of CNC Lathes:

- **Horizontal Lathes:** The workpiece rotates horizontally.
- **Vertical Lathes:** The workpiece rotates vertically.
- **Swiss-Type Lathes:** Ideal for small, intricate parts.
- **Multi-Axis Lathes:** Can move in multiple directions simultaneously.
- **Live Tooling Lathes:** Equipped with rotating tool stations for complex operations.

4. Specifications and Features:

- **Spindle Speed:** Determines how fast the workpiece rotates.
- **Chuck Size:** Refers to the maximum workpiece diameter.
- **Tool Turret:** Holds various cutting tools for automatic tool changes.
- **Bar Feeder:** Loads long bars of material for continuous production.
- **Tailstock:** Supports the workpiece during machining.

5. Applications:

- CNC lathes are widely used in manufacturing, aerospace, automotive, and medical industries.
- They create precision parts such as shafts, bushings, gears, and threaded components.

6. Notable CNC Lathes:

- **TAIG 4029 DSLS Full CNC Lathe:** Entry-level CNC lathe with accurate results and a DSLS system.
- **Baileigh PL-1236E-DRO:** Mid-range lathe with high-quality turning and an industrial-grade digital readout system.
- **Mazak QT-Compact 300MY** and **Haas ST 10Y:** Professional-grade CNC lathes.



Figure: CNC Lathe machine HAAS

ST-30Y is a Y-Axis Turning Centre Lathe Machine. Specifications –

- 3400-rpm Spindle – Belt Drive
- 4000-rpm Live Tooling with C axis, 12-Station BMT65 Turret
- Y-axis milling, drilling, and tapping
- 55-Gallon Coolant Tank
- Travel X, Y and Z Axis
- Number of Tools: 12 Stations
- 30 HP power Features –
- New castings – Increased rigidity and dampening properties
- Improved ergonomics for everyday operation
- Improved chip evacuation
- Increased Z-axis travel
- Increased wedge stance for heavy-duty performance • Wi-Fi standard
- Has Drop wireless file transfer standard
- Safe Run standard
- Has Connect remote monitoring standard
- Two-Stage Automatic Parts Catcher (Optional)
- Sub-Spindle (Optional)
- Auto Door (Optional)



Figure: VF 5 3 axis CNC Machine HAAS

ST-30Y is a Y-Axis Turning Centre Lathe Machine
Specifications–

- Travel X, Y and Z Axis
- 30 HP power
- Spindle Max Speed 8100 rpm
- Coolant Capacity 95 Gallon
- Cooling: Liquid Cooled
- Number of Std T-Slots 7
- 20-Pocket Carousel Tool Changer
- Control Touch Screen

Features –

- More Y-axis travel
- High-power, direct-drive spindle
- Offers more Y-axis travel than the VF-4 models
- Increased wedge stance for heavy-duty performance
- WiFi standard
- Has Drop wireless file transfer standard

- Safe Run standard
- Has Connect remote monitoring standard
- Two-Stage Automatic Parts Catcher (Optional)
- Sub-Spindle (Optional)
- Auto Door (Optional)

Signature

Welding Section

Welding is a fascinating fabrication process that joins two or more parts together by applying heat, pressure, or both.

- Welding fuses materials, creating a strong bond as they cool down.
- It's commonly used for metals and thermoplastics, but it can even be applied to wood.
- The resulting joint is called a **weldment**

Types of Welding Processes:

- **MIG Welding (GMAW):** Uses a wire electrode fed from a spool to create the weld.
- **TIG Welding (GTAW):** Employs a tungsten electrode with a high melting point.
- **Stick Welding (SMAW):** Arc struck between a flux-coated metal rod and the workpiece.
- **Flux Cored Arc Welding (FCAW):** Similar to MIG welding but with a flux-filled wire.
- **Plasma Arc Welding:** Uses a focused plasma arc for precision.
- **Electron Beam Welding:** High-energy electrons create the weld.
- **Laser Beam Welding:** Laser energy melts the materials.
- **Gas Welding:** Combines oxygen and fuel gases for fusion.
- **Automic Hydrogen Welding:** Utilizes hydrogen gas for specific applications.
- **Electroslag Welding:** Ideal for thick materials

Applications:

- Welding is crucial in industries like construction, automotive, aerospace, and shipbuilding.
- It's used to create everything from bridges and pipelines to intricate metal sculptures

working procedure in welding.

1. Preparation:

- **Safety First:** Wear appropriate protective gear, including a welding helmet, gloves, and flame-resistant clothing.

- **Surface Preparation:** Clean the surfaces to be welded. Remove rust, paint, and contaminants.
 - **Joint Design:** Choose the type of joint (e.g., butt joint, lap joint, T-joint) based on the application.
 - **Electrode Selection:** Select the appropriate welding electrode or wire based on the material and process.
2. **Setting Up the Welding Equipment:**
- **Power Source:** Connect the welding machine to a suitable power supply.
 - **Ground Clamp:** Attach the ground clamp to the workpiece.
 - **Electrode Holder:** Insert the welding electrode or wire into the electrode holder.
3. **Arc Initiation:**
- **Arc Welding:** Strike an arc by touching the electrode to the workpiece and then quickly pulling it away.
 - **TIG Welding:** Use a high-frequency start or scratch start to initiate the arc.
4. **Welding Technique:**
- **Maintain Arc Length:** Keep a consistent distance between the electrode and the workpiece.
 - **Travel Speed:** Move steadily along the joint at an appropriate speed.
 - **Weaving:** For wider welds, weave the electrode side to side.
5. **Depositing Filler Metal:**
- **MIG Welding:** The wire electrode automatically feeds into the weld pool.
 - **TIG Welding:** Add filler rod manually to the molten pool.
 - **Stick Welding:** The flux-coated electrode provides both filler metal and flux.
6. **Shielding Gas:**
- **MIG and TIG Welding:** Use inert gases (argon, helium) to protect the weld from atmospheric contamination.
 - **Flux-Cored Arc Welding:** The flux in the wire provides shielding.
7. **Cooling and Solidification:**
- As the weld cools, it solidifies, forming a strong joint.
8. **Post-Weld Cleaning and Inspection:**
- **Remove Slag:** For stick welding, chip off the slag.
 - **Visual Inspection:** Check for defects (cracks, porosity, incomplete fusion).
 - **Grinding and Finishing:** Smooth any rough edges.
9. **Safety Measures After Welding:**
- Allow the weld to cool completely.
 - Inspect the finished weld for quality.

Signature

Fabrication Section

Fabrication is the construction of items from different parts using at least one of a range of processes and materials such as metal, laminates, wood and other solid surface materials.

Frequently used in relation to metal fabrication, the steps involved can include stamping, welding, cutting, bending, and assembling processes.

As with other manufacturing processes, fabrication can be done manually, but the processes are often automated these days with the use of computer aided designs (CAD) that can be programmed into computer numerical control (CNC) technologies that can communicate directly with machines on the shop floor, reducing lead times, costs and material usage while improving accuracy and quality.

The fabricator's process tends to make items from manufactured materials like steel, that can then be assembled to create larger metal structures. In addition, fabrication involves making components for items such as engines, machines, tools, and household appliances.

Fabrication uses semi-finished or raw materials to make something from start to finish, as opposed to simply assembling it. This work is typically completed by a fabrication shop (or fab shop) that will bid on the contract for the job based on engineering drawings and specifications.



Combined Shearing Machine



Sheet Cutting Shearing Machine



Vertical Drill



Bending Machine



Rolling Machine



Radial Arm Drill

Signature

Day 02

TTI



Fig: VF-6 machine

VF-6 machine specification:

Maximum capacity 1800 kg

X-axis travel -1626mm

Y-axis travel -813mm

Z-axis travel -647mm

X-axis Table size -1620mm

Y-axis Table size -705mm

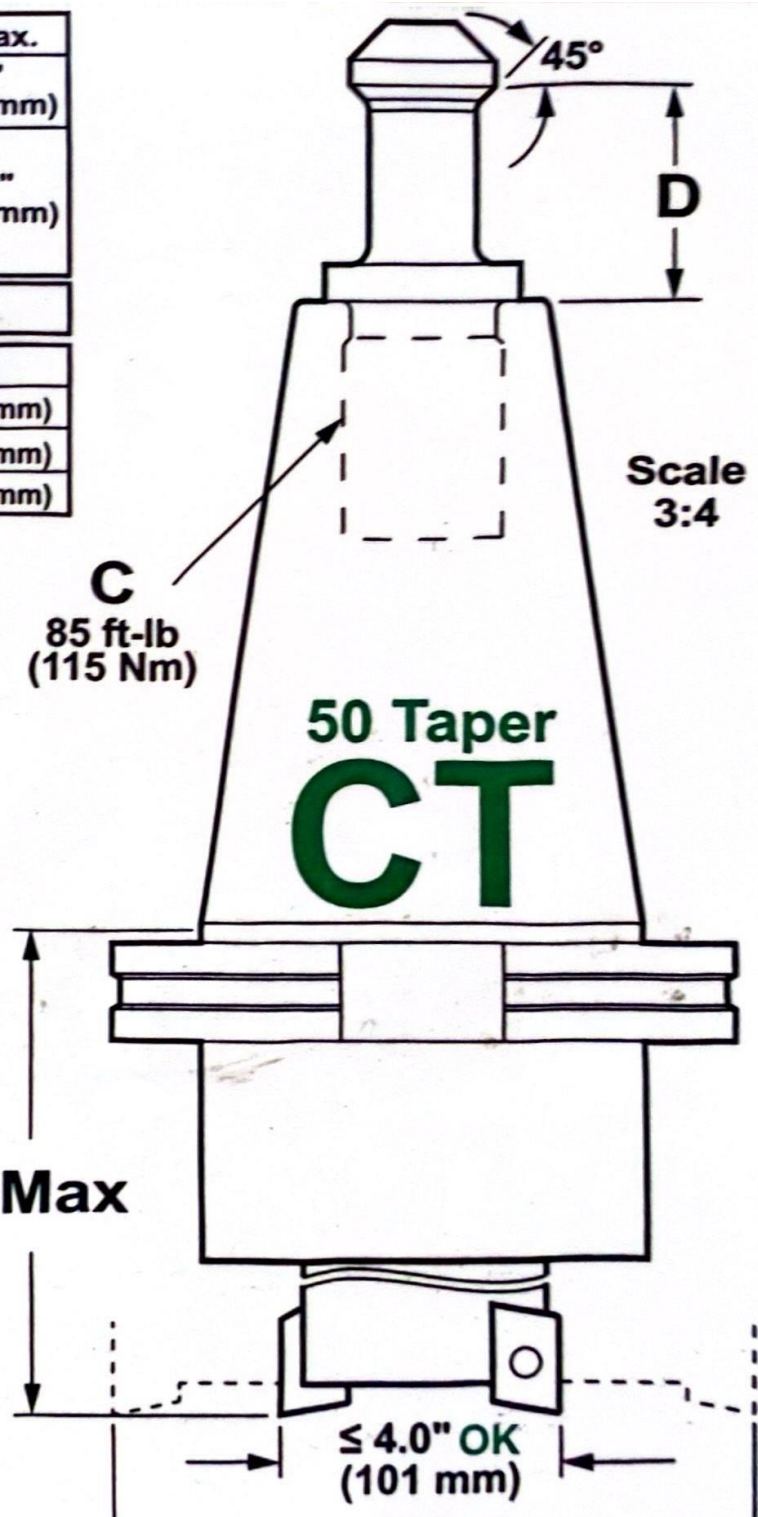
Spindles-8100 rpm

Tool Changers - 30+1 Side-Mount Tool Changer

Coolant Management

- Variable Flow Coolant
- Programmable Coolant Nozzle
- Chip Auger
- Window Air Blast

MACHINE	A Max.	B Max.
VF-3YT, VF-5	16" (406 mm)	7" (178 mm)
VF 6-12		10" (254 mm)
EC 1600	20" (508 mm)	
30lb Max. (13.6kg Max.)		
50 Taper	C	D
BT	M24X3	1.386" (35.2 mm)
CT	1-8	1.386" (35.2 mm)
DIN/ISO	M24X3	1.386" (35.2 mm)



VF-2 Machine:

X-axia travel -762mm

Y-axia travel -408mm

Z-axia travel -503mm

X-axia Table size -910mm

Y-axis Table size -400mm

Signature

Automobile Section

There are a wide variety of propulsion systems available or potentially available for automobiles and other ...vehicles. Options included internal combustion engines fueled by petrol, diesel, propane, or natural gas; hybrid vehicles, plug-in hybrids, fuel cell vehicles fueled by hydrogen and all electric cars. Fueled vehicles seem to have the advantage due to the limited range and high cost of batteries. Some options required construction of a network of fueling or charging stations. With no compelling advantage for any particular option, car makers pursued parallel development tracks using a variety of options. Reducing the weight of vehicles was one strategy being employed. The use of high-technology (such as electronic engine control units) in advanced designs resulting from substantial investments in development research by European countries and Japan seemed to give an advantage to them over Chinese automakers and parts suppliers who, as of 2013, had low development budgets and lacked capacity to produce parts for high-tech engine and power train designs

Gasoline Engine Components:

- Piston
- Gear
- Cam
- Camshaft
- Intake valve

- Exhaust valve
- Spark plug
- Connecting rod

Pistons, also known as “little powerhouses,” are small but mighty components that play a crucial role in many mechanical systems, from car engines to industrial pumps.



Fig: Piston

The camshaft is critical to the basic function of an engine. Comprised of two distinct parts, the cams and the shaft, the camshaft is the element that enables valves to open. As the shaft rotates, the egg-shaped cams (or "lobes") push the valves open in sync with the crankshaft gear.

Signature

Day 03

Heat Treatment

Heat treatment is a controlled process used to alter the metallurgical property of materials. Heat treatment basically does metal testing, heat, soaking, cooling, hardness testing, tempering type works. Heat treatment basically deals with different type of materials. Such as- Mild steel, Alloy steel etc. Steel materials are first carburized and then hardened. To do carburizing, first carburizing compound is made with

1. Wood Coal- 58%
2. Sodium- 02%
3. Barium -15%
4. Stone coal- 25%
5. Molasses according to need



Figure: Carburizing Compound (Left) & Carburizing Box (Right)

Carburizing temprature 950°C-1000°C. Semi muffle gas fire furnace is used to do the carburizing. Electric super heating furnace is used for hardening at 750°C-800°C. Salt bath muffle gas fire furnace is also used in carburizing & hardening purpose. Block oil hardness testing machine is used for hardness testing. To cool down the job air, oil and water is used.

Hand Press is used to straighten the jobs. Heat treatment types are-

1. Annealing temprature 600°C-650°C
2. Normalizing temprature 600°C-650°C
3. Hardening temprature 750°C-800°C

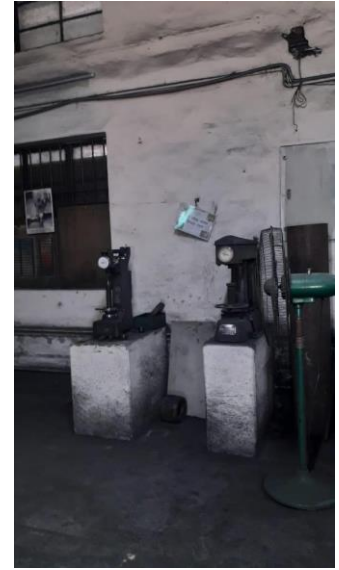
4. Tempering temprature 150°C-550°C



(a)



(b)



(c)

Figure (a): Muffle furnace Figure (b): Hand Grinding Wheel Figure(c): Rockwell Hardness Testing Machine

Signature

Foundry Shop

Machines available at Foundry Shop:

1. Dumping Barrel / SandBlaster Machine
2. Sand Muller Machine
3. Sand Mixture Machine
4. Grinding Machine
5. Pour Oven (Heating Capacity: 350°C-400°C)

6. Cupola Furnace (Capacity: 1-1.5ton, used for Cast Iron)
7. Tilting Furnace (Capacity: 300kg, used for Copper, Aluminum)
8. Crucible Furnace (Capacity: 40kg-100kg, used for Copper, Aluminum, Cast Iron, White Metal)
9. Induction Furnace (Capacity 300kg, it's fully hydraulic, used for Cast Iron, Mild Steel, Stainless Steel)
10. Ladle.
11. Crucible

The main purpose of a foundry shop is to cast any metal according to desired shape. Usually cast iron is used for producing automobile parts and aluminum is used for producing aero plane parts.

There are two types of casting:

- Sand Casting
- Die Casting

First, a pattern is needed for casting. Wood is the best option for making patterns. Then mold is needed to be created using sand. Green sand can be used for mold making where Graphite and Bentonite is added to increase stickiness. It is possible to get better finishing in die casting rather than sand casting. In die casting Hot Die Steel or Cast Iron is used for making die. For casting Cast Iron materials, sand mixing ratio should be around (using CO2 process):

- Sand 90%
- Sodium Silicate 7%
- Coal Dust 1.5%
- Saw Dust 1.5%

On the other hand, for casting Stainless Steel, Mild Steel, Copper, Aluminum materials, sand mixing ratio should be around (using CO2 process):

- Sand 92%
- Sodium Silicate 6%
- Saw Dust 2%

The foundry graded sand should have circular grain. Otherwise, they will not stay together while making molds.



Fig: Different foundry operations and their machinery



Fig: Mold making

Signature

Pattern Shop

A pattern is a shaped form of wood or metal around which sand is packed in the mold. When the pattern is removed the resulting cavity is the exact shape of the object to be cast. The pattern must be designed to be easily removed without damage to the mold. It must be accurately dimensioned and durable enough for the use intended. Either one-time use or production runs. Pattern materials:

Wood - common material because it is easy to work, but it warps

Metal - more expensive to make, but lasts much longer Plastic
compromise between wood and metal.



Figure: Gamari Wood

Types of Pattern

1. *One Piece/Solid Pattern*

Used for simple shape low quantity production. Made of wood and these are expensive.

2. *Split Patterns*

Two-piece patterns are made such that each part forms a portion of the cavity for casting.

3. *Match-plate pattern*

Two pieces were constructed by securing each half on the opposite sides of a single plate.

4. *Cope & Drag Pattern*

Two-piece were constructed by securing each half on the cope and Drag For making a pattern, we need various tools. They are given below:

- Steel tape

- Steel roll
- Cut pencil
- Standard Scale
- Scriber
- Tri-square
- Hand jaw
- Hand planer
- Scissor
- Divider



Figure: Pattern Shop Tools



Figure: Standard Scale

Band saw machine



Figure: Band Saw Machine

A band saw is a power tool that uses a blade consisting of a continuous band of metal with teeth along one edge to cut various work pieces. The band usually rides on two wheels rotating in the same plane, although some band saws may have three or four wheels. Band sawing produces uniform cutting action as a result of an evenly distributed tooth load. Band saws are used for woodworking and metalworking, for cutting a variety of other materials. They are particularly useful for cutting irregular or curved shapes but can make straight cuts. The minimum radius of a curve that can be cut on a particular saw is determined by the width of the band and its kerf.

Signature

Day 04

Molding

Surface Grinding Machine:

Grinding is a metal cutting operation like any other process of machining removing metal in comparatively smaller volume. The cutting tool used is an abrasive wheel having many numbers of cutting edges. The machine on which grinding the operation is performed is called a grinding machine. This machine is used to smooth the surface. There is a magnetic table in this machine

which can be fixed and can be moved up, down and left and right. The rpm of the cutting wheel is 2500. The smallest dimension it can create is 0.0001 and the largest dimension is 0.001.

Copying Milling Machine:

Copy milling is more commonly known as free-form milling. As the name would suggest, a copy is made of the object being milling. The cutter generates complex geometric (free) forms which are typically non-angular. The machining technique is commonly used in the tool and mold-making industry. There are 4 different types of cutters used in this machine: End mill cutter, Single lift cutter, Ball nose cutter , Point Depth cutter.

Tool set:

1. Template
2. Blind job
3. Stylers
4. Single lip cutter
5. Table vice
6. Ball Pein hammer
7. Adjustable wrench
8. Vernier micrometer
9. Nut bolt

Industrial Engineering Division

The Industrial Engineering Division in Bangladesh Industrial Technical Assistance Centre (BITAC) is responsible for analyzing and improving the production processes in order to increase efficiency and reduce costs. The division may have a number of responsibilities, including:

- Process improvement: The division is responsible for identifying areas for improvement in the production process and implementing changes to increase efficiency, reduce costs, and improve product quality.

- Time and motion studies: The division is responsible for studying the time and motion required to perform specific tasks in the production process, and identifying ways to reduce the time and improve the efficiency of these tasks.
- Cost analysis: The division is responsible for analyzing the costs associated with the production process, such as labor, materials, and equipment, and identifying ways to reduce these costs.
- Capacity planning: The division is responsible for analyzing the facility's production capacity and identifying ways to increase capacity in order to meet production needs.
- Workplace design: The division is responsible for designing the facility and workstations to optimize productivity, safety and ergonomic conditions.
- Production planning and control: The division is responsible for planning and controlling the production process to ensure that products are produced on time and to the required standards.
- Statistical process control: The division is responsible for implementing statistical process control methods to monitor and improve the quality of the production process.
- Training: The division is responsible for providing training for employees on new processes and equipment, as well as on safety and ergonomic issues.

The Industrial Engineering Division in BITAC plays a key role in ensuring that the facility is operating efficiently and effectively and that the products produced meet the required standards and are delivered on time to customers.

Signature

Production Control Division

The Production Control Division in Bangladesh Industrial Technical Assistance Centre (BITAC) is responsible for managing and coordinating the production process to ensure that the facility is operating efficiently and effectively. The division may have a number of responsibilities, including:

- **Scheduling:** The division is responsible for creating and managing the production schedule, ensuring that all necessary equipment, materials, and personnel are available when needed.
- **Quality control:** The division is responsible for monitoring the quality of the products produced by the facility and implementing quality control procedures to ensure that the products meet the required standards.
- **Inventory management:** The division is responsible for managing the facility's inventory of materials and supplies, ensuring that there is always enough on hand to meet production needs.
- **Cost control:** The division is responsible for monitoring and controlling the costs associated with the production process, such as labor, materials, and equipment. **Safety:** The division is responsible for ensuring that all safety procedures are in place and being followed in the facility to protect the workers, equipment, and the environment.
- **Continuous improvement:** The division is responsible for identifying areas for improvement in the production process and implementing changes to increase efficiency and reduce costs.
- **Communication:** The division is responsible for communicating with other departments and divisions within the facility to ensure that all necessary information is shared and that production runs smoothly.
- **Record Keeping:** The division is responsible for keeping records of production data such as production schedule, inventory, and quality control data, that can be used for analysis and reporting.

Overall, the Production Control Division in BITAC plays a key role in the smooth operation of the facility and in ensuring that the products produced meet the required standards and are delivered on time to customers.

Signature

Conclusion

This file provided us with a wealth of useful information on several departments. We gained knowledge of inspection, mold, pattern shop, foundry shop, heat treatment, fabrication, and machine maintenance in addition to all kinds of conventional and nonconventional machinery. We also learned how PCD controls production and how the IE department handles BITAC's overall cost of work. Our theoretical understanding has been enhanced by this practical understanding. Every BITAC department was incredibly cordial. Every lecturer had a strong love for what they taught us. We found this file to be very helpful, and it will continue to be so in the future.